

Subject Description Form

Subject Code	EIE3312
Subject Title	Linear Systems
Credit Value	3
Level	3
Pre-requisite/ Co-requisite/ Exclusion	AMA2111 Mathematics I
Objectives	1. To provide students with basic concepts and techniques for the modelling and analysis of linear continuous-time and discrete-time signals and systems. 2. To provide students with an analytical foundation for further studies in Communication Engineering and Digital Signal Processing.
Intended Subject Learning Outcomes	Upon completion of the subject, students will be able to: <u>Category A: Professional/academic knowledge and skills</u> 1. Understand the representations and classifications of the signals and systems. 2. Understand the modelling of linear systems. 3. Use different techniques to analyze and design systems. 4. Apply software tools to laboratory exercises for experimenting with theories, and to the analysis and design of signals and systems. 5. Appreciate the advantages and disadvantages of using the different representations and modeling approaches. <u>Category B: Attributes for all-roundedness</u> 6. Present ideas and findings effectively. 7. Think critically and learn independently. 8. Work in a team and collaborate effectively with others.
Subject Synopsis/ Indicative Syllabus	Syllabus: 1. <u>Signal Representation</u> Signal Classification, Continuous and Discrete-Time Signals, Random Signals. Time-Domain and Frequency-Domain Representations. 2. <u>Continuous-Time and Discrete-Time Systems</u> Impulse Representation and Convolution, Linear Time-Invariant Systems. Properties of Systems: Causality, Time Invariance, Linearity, Systems with Memory, Inverse of a System, Stability. LTI Systems: Differential and Difference Equation Representation, Block Diagram Representations. 3. <u>Fourier Representations for Signals</u> Reviews on Periodic and Nonperiodic Signals, Continuous and Discrete Signal, Fourier Series and Transform, Frequency Spectra. Properties of Fourier Representations, Time Functions, Applications on System Frequency Response and Signal Frequency Spectrum. Frequency Response of LTI Systems, Sampling. Discrete-Time Fourier Transform, 4. <u>Laplace Transform</u>

[illegible]

	Explanation of the appropriateness of the assessment methods in assessing the intended learning outcomes:	
	Specific Assessment Methods/Tasks	Remark
	Short quizzes	These measure the student's understanding of the theories and concepts as well as their comprehension of the subject materials.
	Assignments and Examination	End-of-chapter-type problems are used to evaluate the student's ability to apply concepts and skills learned in the classroom. Students need to think critically and learn independently to come up with an alternative solution to an existing problem.
	Laboratory sessions	Each student is required to produce a written report. The accuracy and presentation of the report will be assessed.
Student Study Effort Required	Class contact (time-tabled):	
	• Lecture	24 Hours
	• Tutorial/Laboratory/Practice Classes	15 hours
	Other student study efforts:	
	• Lecture: preview/review of notes; homework/assignment; preparation for test/quizzes/examination	36 Hours
	• Tutorial/Laboratory/Practice Classes: a preview of materials, revision, and/or reports writing	30 Hours
	Total student study effort:	105 Hours
Reading List and References	Reference Books: [1] Ed. Kamen and Bonnie Heck, <i>Fundamentals of Signals and Systems Using the Web and Matlab</i> , 3/e, Prentice-Hall, 2007. [2] M.J. Roberts, <i>Fundamentals of Signals & Systems</i> , McGraw-Hill, 2008 [3] Simon Haykin and Barry Van Veen, <i>Signals and Systems</i> , Wiley, 2003. [4] Charles L. Phillips, et al., <i>Signals, Systems, and Transforms</i> , 5/e, Prentice-Hall, 2014.	